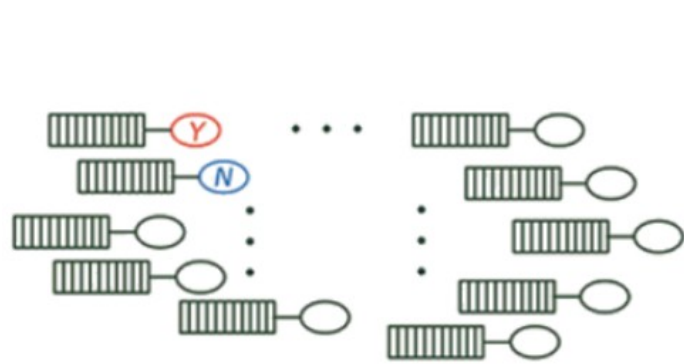


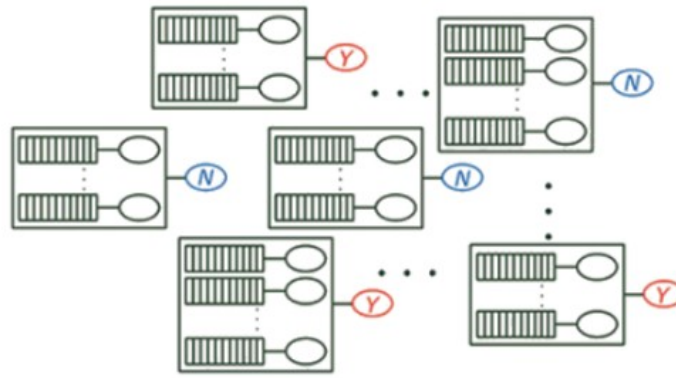
Automatic Image Analysis

Weakly-Supervised Learning

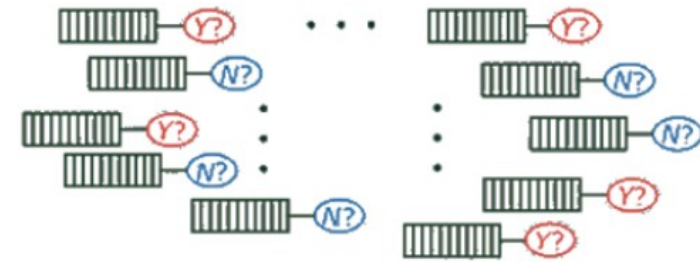
Types of Weak Supervision



(a) Incomplete supervision



(b) Inexact supervision



(c) Inaccurate supervision

Incomplete Supervision

- Two types
 - Active learning: an “oracle” can be queried to get ground-truth labels for selected unlabeled instances
 - Semi-supervised learning: automatically exploit unlabeled data
 - Transductive learning:
 - ”Closed-world” assumption:
Test data are given in advance.
Goal: optimize on test data
 - Pure semi-supervised learning:
 - ”Open-world” assumption:
Test data are unknown.
Unlabeled data are not necessarily test data

Transductive Learning

- Phenomenon f maps inputs (instances) $\mathbf{x}$ to outputs (labels)
 $y = f(\mathbf{x})$ (here $y \in \{-1, 1\}$)
- Given a set of labeled examples $\{(\mathbf{x}_i, y_i): i=1, \dots, n\}$
- and a set of unlabeled examples $\mathbf{x}'_1, \dots, \mathbf{x}'_m$
- the goal is to directly infer its labels $y'_1, \dots, y'_m$
- So there is no need to construct a function f .
- Output is a vector of labels, not the function f .
- Transfer the information from labeled examples to unlabeled.
- c.f. Vapnik: Support Vector Machines

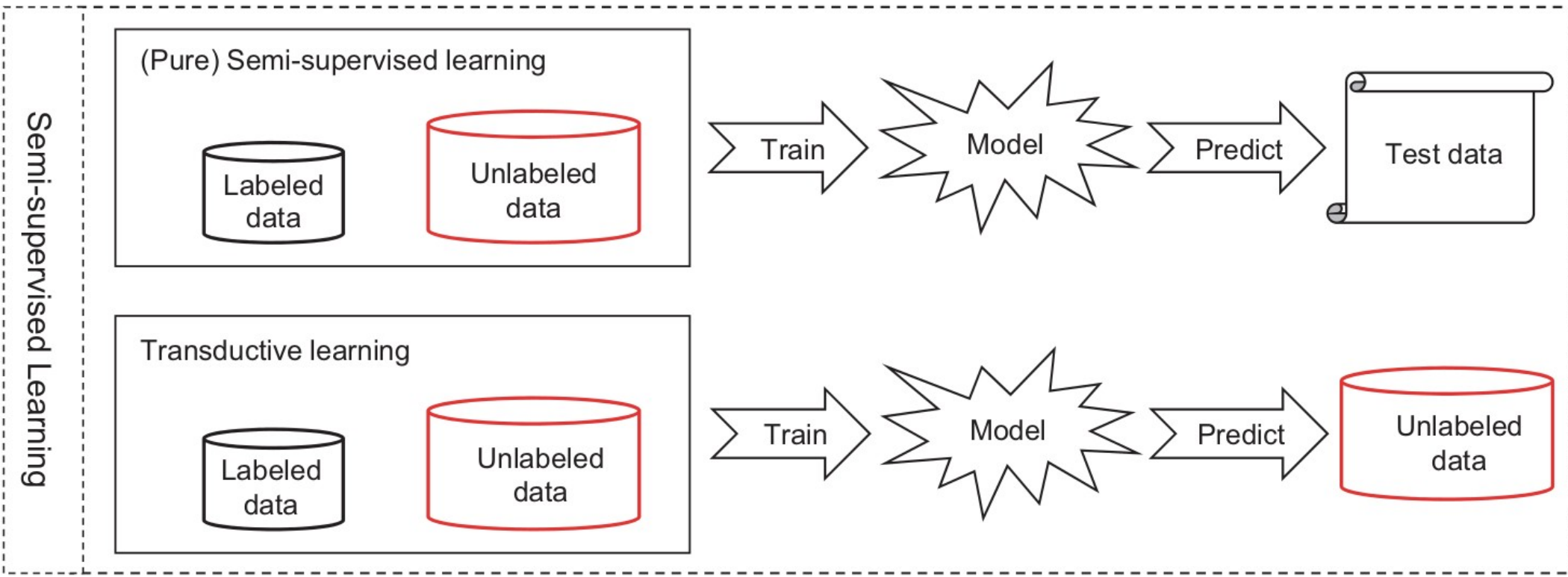
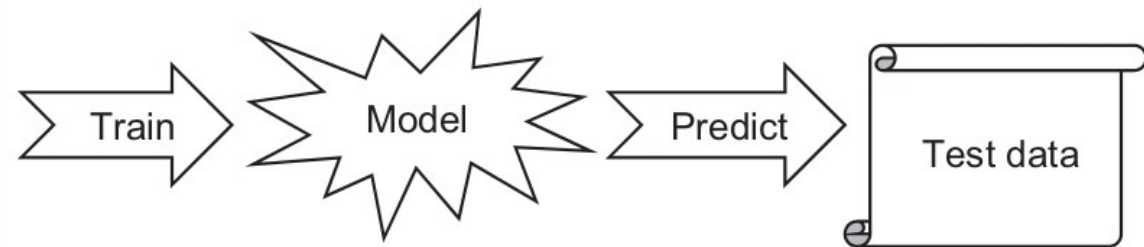
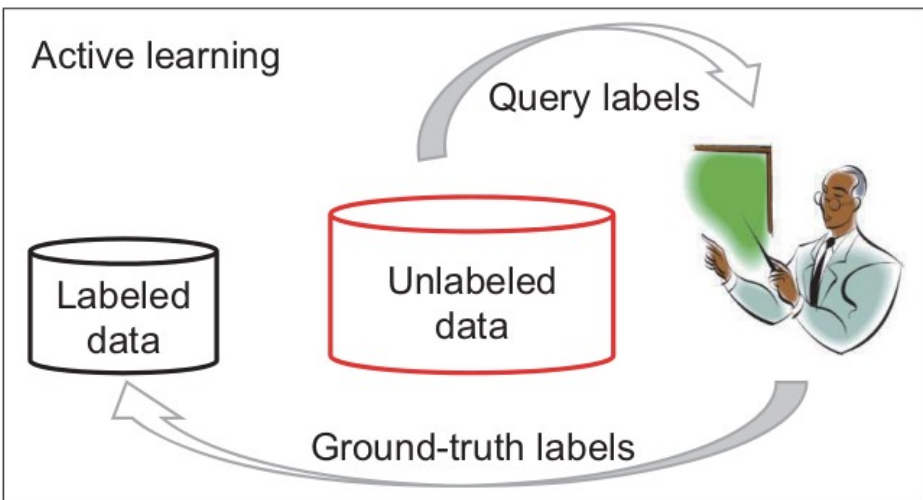
Three Learning Tasks

- Induction: $\{(\mathbf{x}_i, y_i) : i = 1, \dots, n\} \mapsto f$
- Induction with unlabeled data: $\{(\mathbf{x}_i, y_i) : i = 1, \dots, n\} \cup \{x'_1, \dots, x'_m\} \mapsto f$
- Transduction: $\{(\mathbf{x}_i, y_i) : i = 1, \dots, n\} \cup \{x'_1, \dots, x'_m\} \mapsto (y'_1, \dots, y'_m)$.

The choice will depend on

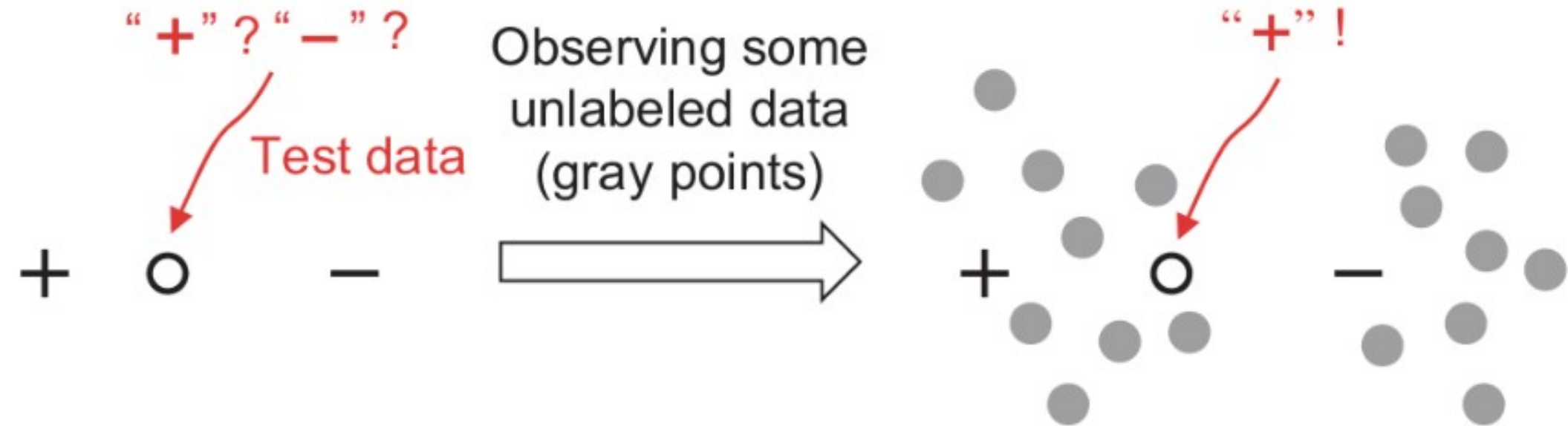
- Availability of unlabeled data
- Need for interpretability
- Time considerations

Active, Pure Semi-Supervised and Transductive Learning



Usefulness of Unlabeled Data

- Without human intervention
- Cluster assumption



Modeling Classification with Unlabeled Data

- Mixture model for binary classification of n samples

$$h(\mathbf{x}) = \arg \max_{c \in \{Y, N\}} \sum_{j=1}^n P(y_i = c | g_i = j, \mathbf{x}_i) \\ \times P(g_i = j | \mathbf{x}_i),$$

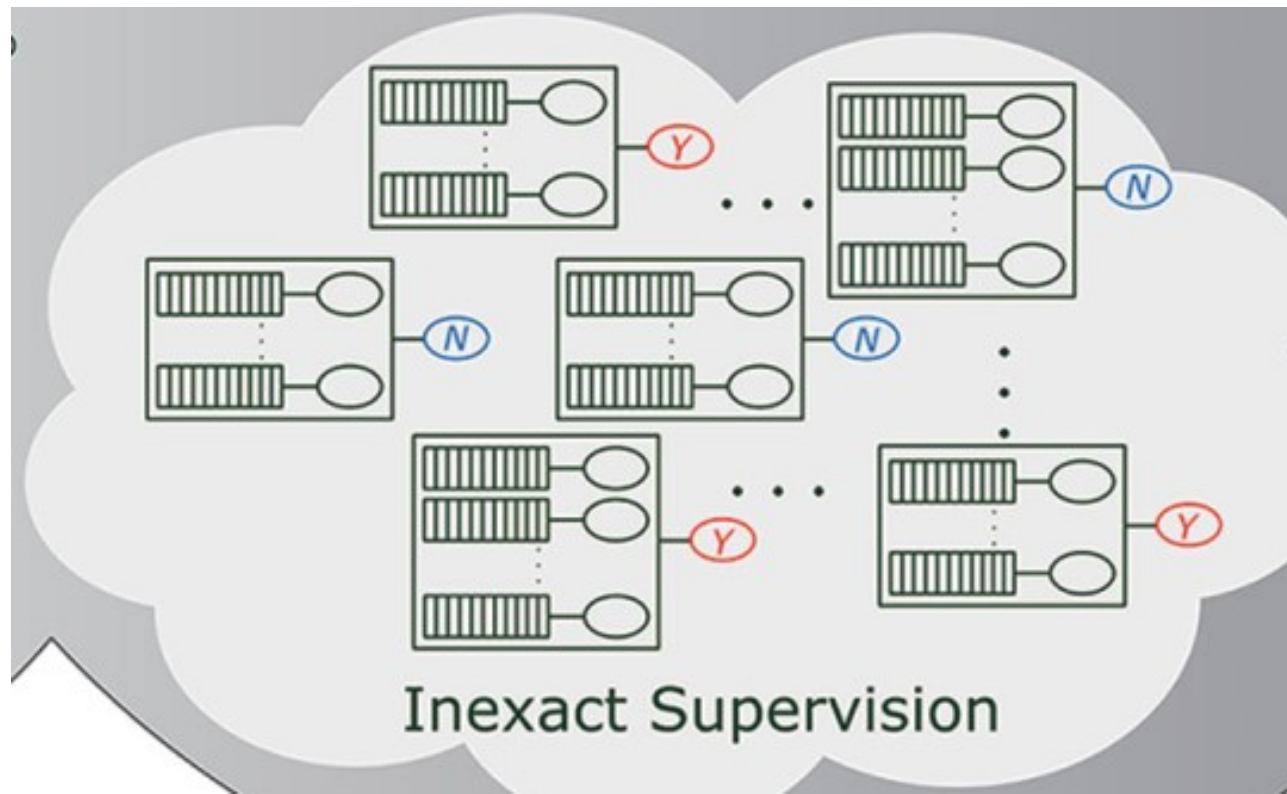
where

$$P(g_i = j | \mathbf{x}_i) = \frac{\alpha_j f(\mathbf{x}_i | \theta_j)}{\sum_{k=1}^n \alpha_k f(\mathbf{x}_i | \theta_k)}.$$

- Only the first term requires labeled information

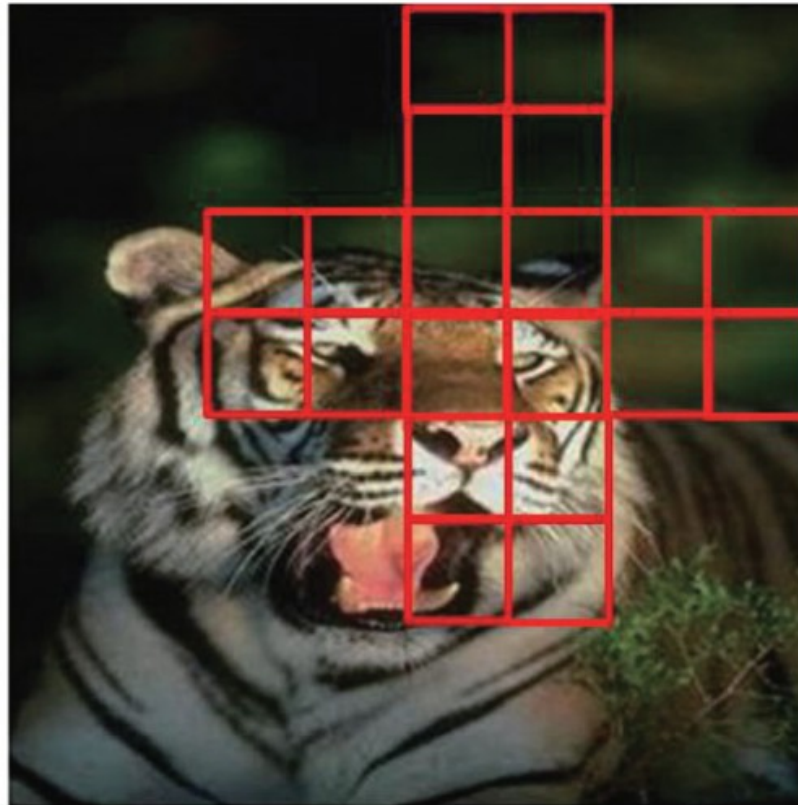
Inexact Supervision

- Some supervision information is given. But not as exact as desired.
- In Multi-Instance Learning (MIL)
 - training data is provided in a “bag” (set) with a label, if one of the instances in the bag has the label.



Inexact Supervision

- Image (bag) in binary classification (Y/N) problem
 - contains many patches (instances).
 - The image is labeled positive, if at least one patch corresponds to the class
 - Goal: Predict labels for unseen bags.



Inaccurate Supervision

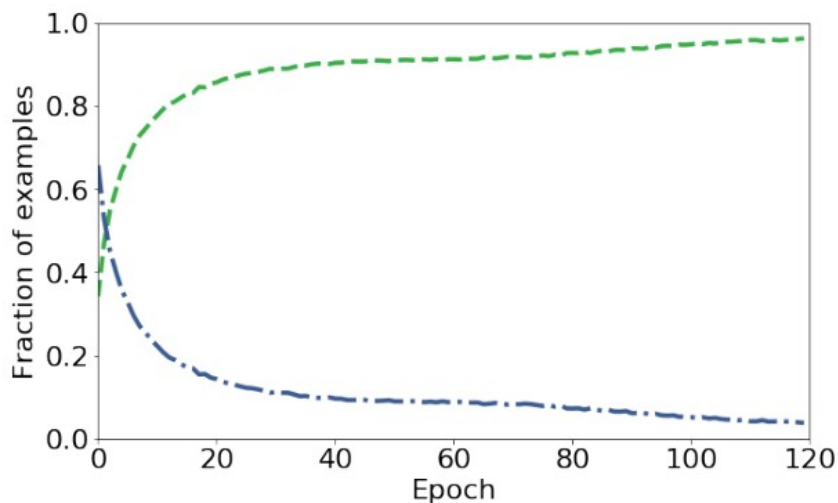
- Supervision information is not always ground truth.
- Two examples:
 - Early learning regularization against noisy labels
 - Negative learning against noisy labels

Early-Learning Regularization

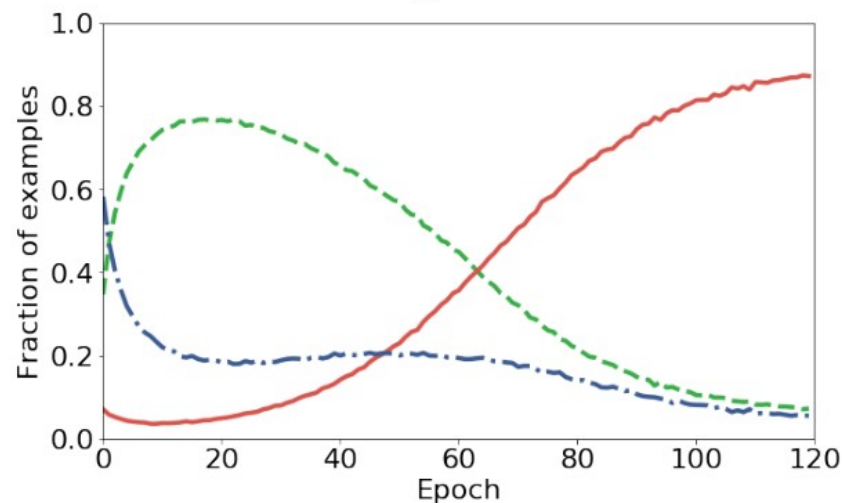
- Given: training data with noisy labels
 - i.e. some labels are wrong
- If the network is very “competent”, it memorizes all the training data correctly.
- However, if some of the training labels are wrong, exactly this should not happen.
- Is there anything “special” about noisy labels that could be used to identify them automatically?
- Liu, S., Niles-Weed, J., Razavian, N., & Fernandez-Granda, C. (2020). Early-learning regularization prevents memorization of noisy labels. *Advances in neural information processing systems*, 33, 20331-20342.

Training Results for Clean and Wrong Labels

Clean labels

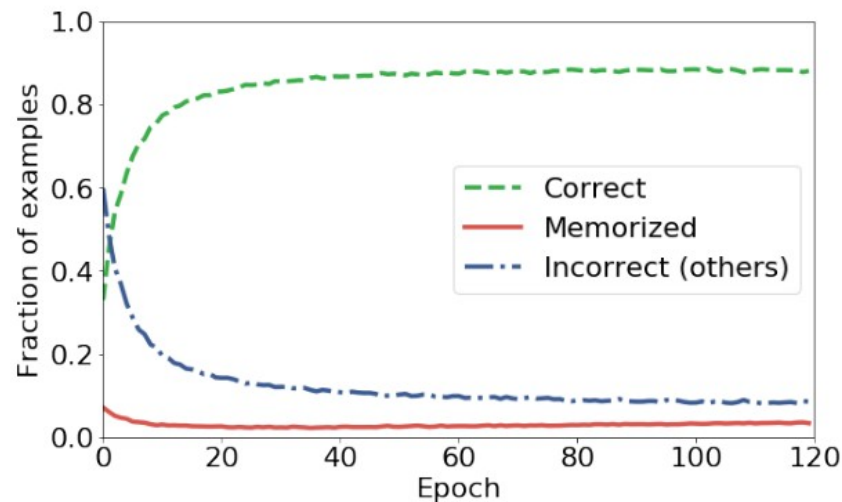
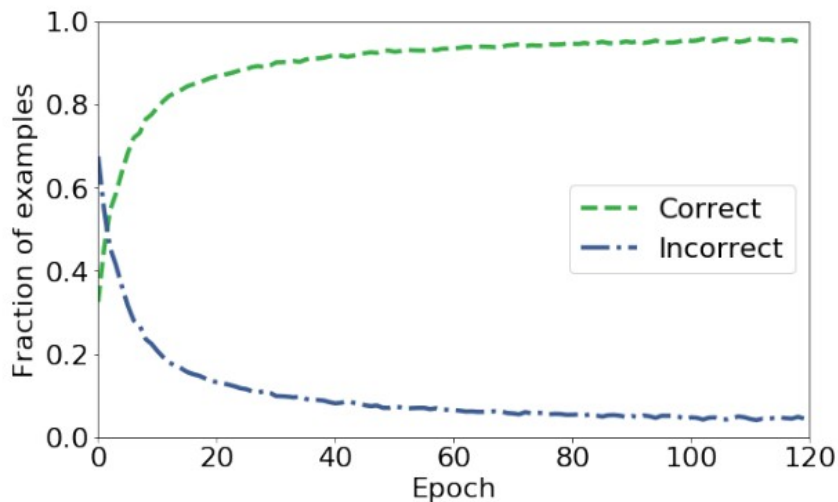


Wrong labels



Cross Entropy

Early-learning
Regularization



- Memorization: wrong labels are learned

Experiment with Simulated Data

- Mixture of two Gaussians
with means separated by opposing unit vectors

$$\mathbf{x} \sim \mathcal{N}(+\mathbf{v}, \sigma^2 I_{p \times p}) \quad \text{if } \mathbf{y}^* = (1, 0)$$

$$\mathbf{x} \sim \mathcal{N}(-\mathbf{v}, \sigma^2 I_{p \times p}) \quad \text{if } \mathbf{y}^* = (0, 1)$$

- Hyperplane through origin is optimal separator.
- Wrong label with probability Δ

$$\mathbf{y}^{[i]} = \begin{cases} (\mathbf{y}^*)^{[i]} & \text{with probability } 1 - \Delta \\ \tilde{\mathbf{y}}^{[i]} & \text{with probability } \Delta, \end{cases}$$

- Use of cross entropy loss
- Results
 - Initially, the accuracy on wrongly labeled examples increases.
 - Over time, the importance of the wrong labels grows.

Result

- **Theorem** (informal):
- For any $\Delta \in (0, 1)$
under fulfillment of further conditions
there exists an iteration threshold T such that
 - **Early learning succeeds**: For $t < T$, the classifier has **higher accuracy** on **wrongly labeled** examples than at initialization (considering **correct labels** of the **wrongly labeled**).
 - Learning **gradients from correct examples vanish** between $t=0$ and $t=T$
 - **Memorization occurs** as $t \rightarrow \infty$

Early-Learning Regularization

- Cross entropy loss (**y**: training label, **p**: estimate)

$$\mathcal{L}_{\text{CE}}(\Theta) := -\frac{1}{n} \sum_{i=1}^n \sum_{c=1}^C \mathbf{y}_c^{[i]} \log \mathbf{p}_c^{[i]}$$

- Stochastic gradient descent eventually results in memorization.
- **Target** vector **t** for each example computed from past outputs.

- New early-learning regularization loss

$$\mathcal{L}_{\text{ELR}}(\Theta) := \mathcal{L}_{\text{CE}}(\Theta) + \frac{\lambda}{n} \sum_{i=1}^n \log \left(1 - \langle \mathbf{p}^{[i]}, \mathbf{t}^{[i]} \rangle \right)$$

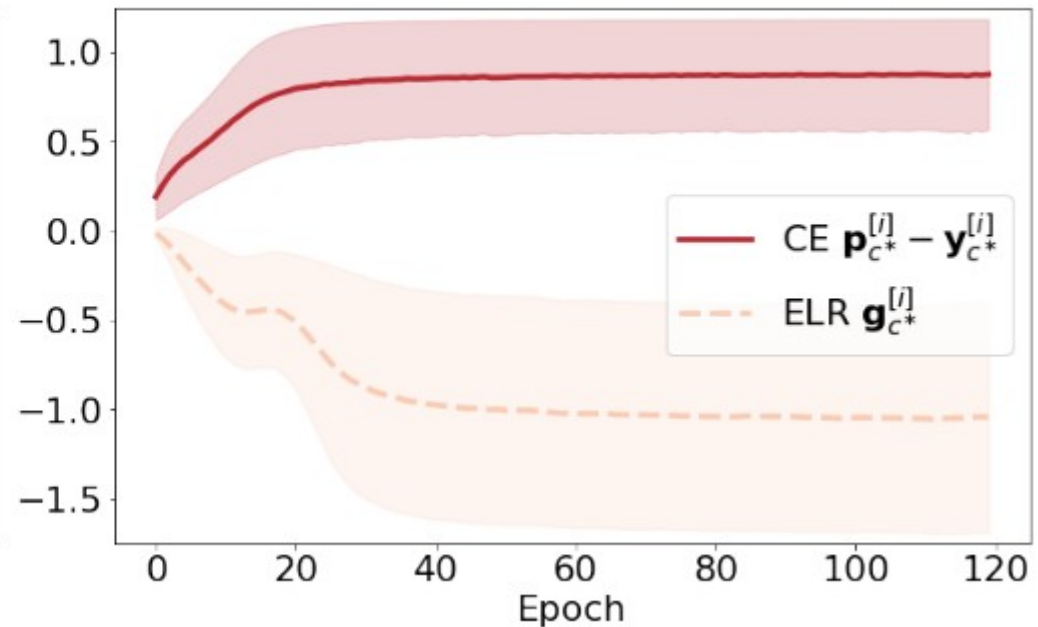
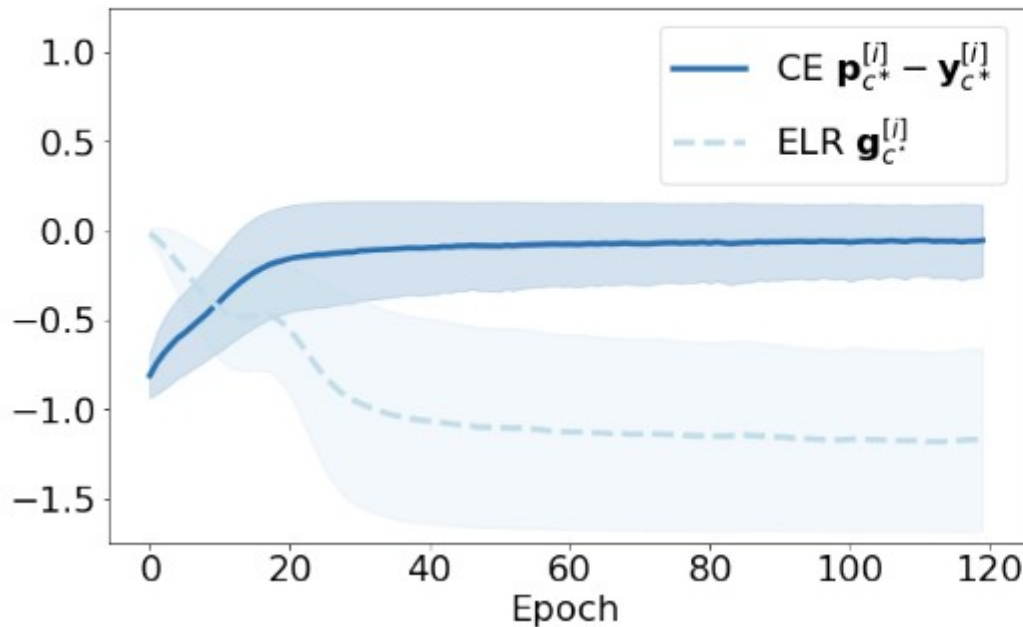
$\langle \cdot, \cdot \rangle$: dot product (similarity)

- **Target** estimation may remain a research issue.

Effect of Early Learning Regularization on Loss Gradient

Clean labels

Wrong labels



- Gradient is sum of CE and ELR.
- Early, cross entropy CE of clean labels dominates.
- Later, ELR eliminates effect of wrong labels and maintains dominance of clean labels.

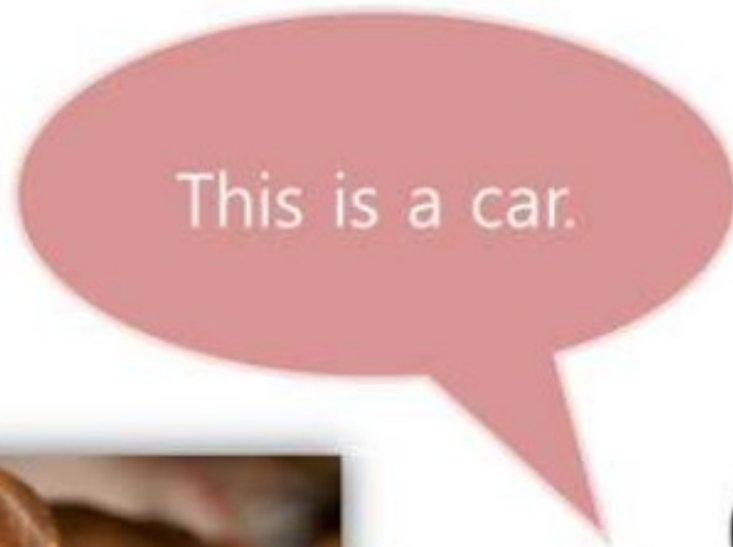
Negative Learning

- Negative learning for noisy labels:
 - Kim, Y., Yim, J., Yun, J., & Kim, J. (2019). NLNL: Negative learning for noisy labels. In Proceedings of the IEEE/CVF international conference on computer vision (pp. 101-110).

Positive vs. Negative Learning

- **Positive Learning (PL)**
 - “input image belongs to this label”
 - classical method for training
- Problematic, if there are “noisy” labels
 - The label may be wrong.
 - Training then causes wrong decision boundaries,
 - as the wrong labels are memorized.
- **Negative Learning (NL)**
 - “input image does not belong to any of the labels it does not carry”
- Less problematic, if there are “noisy” labels

Positive vs. Negative Learning



Given noisy label : Car



Positive vs. Negative Learning

- **Positive Learning (PL)**

- with cross entropy loss

$$\mathcal{L}(f, y) = - \sum_{k=1}^c y_k \log p_k$$

- **Negative Learning (NL)**

- loss

$$\mathcal{L}(f, \bar{y}) = - \sum_{k=1}^c \bar{y}_k \log(1 - p_k)$$

- selected randomly from the labels of all classes except for the given label y

Complementary Label Generation

Algorithm 1 Complementary label generation

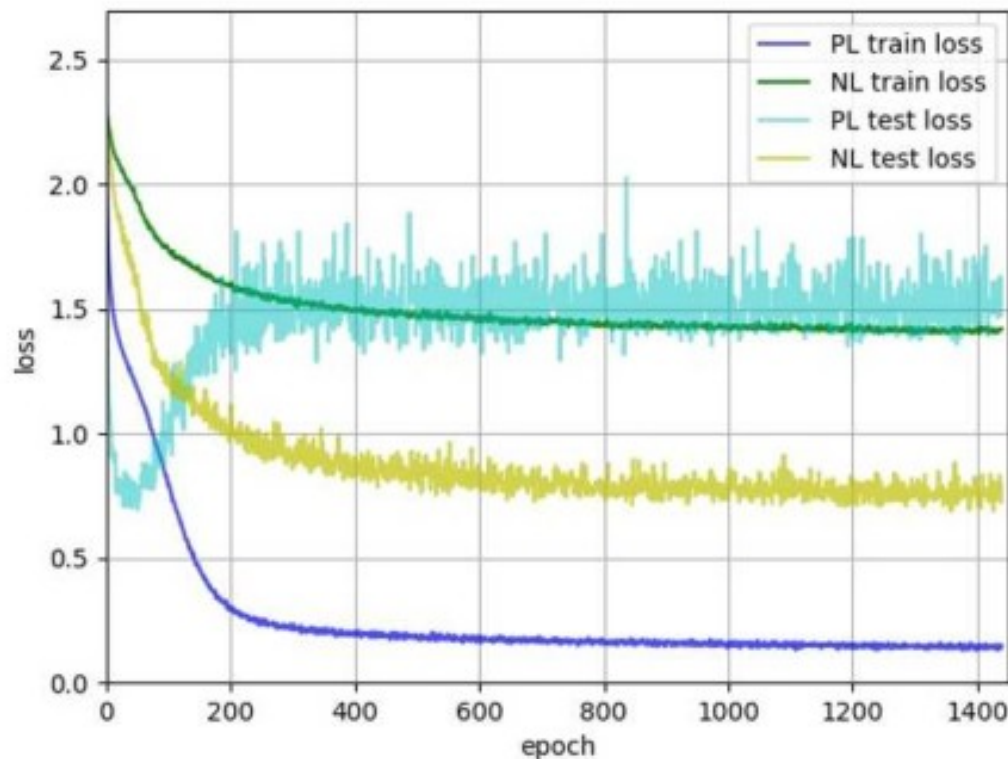
Input: Training label $y \in \mathcal{Y}$

while iteration **do**

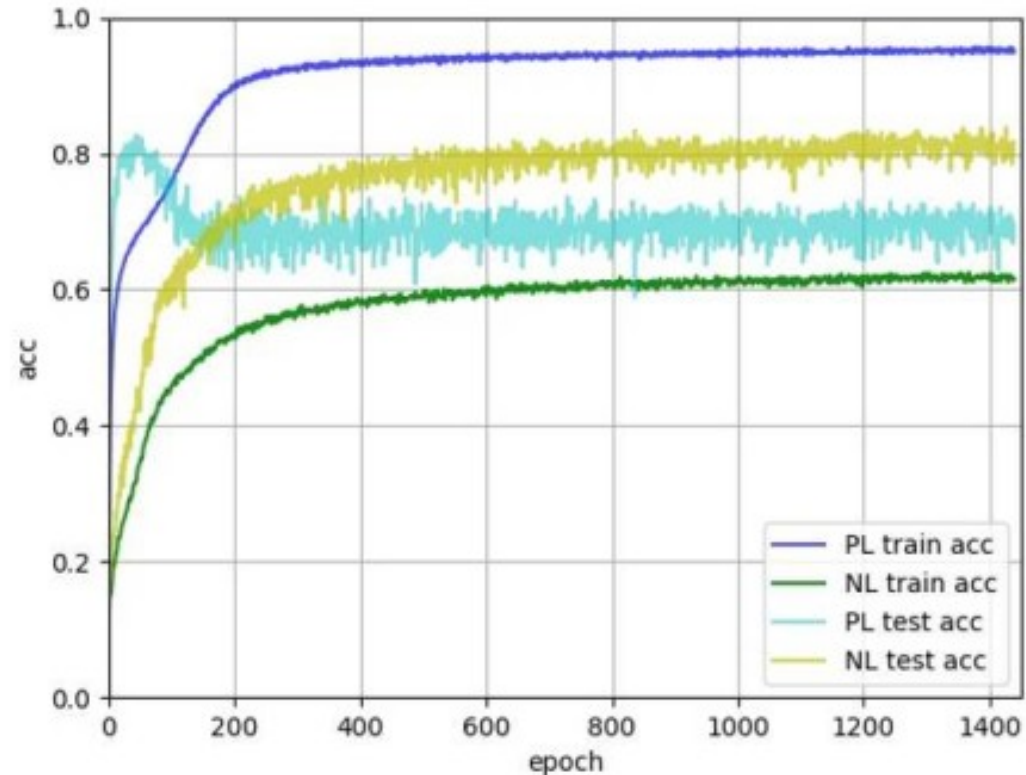
$\bar{y}$ = Randomly select from $\{1, \dots, C\} \setminus \{y\}$

Output: Complementary label $\bar{y}$

Positive vs. Negative Learning



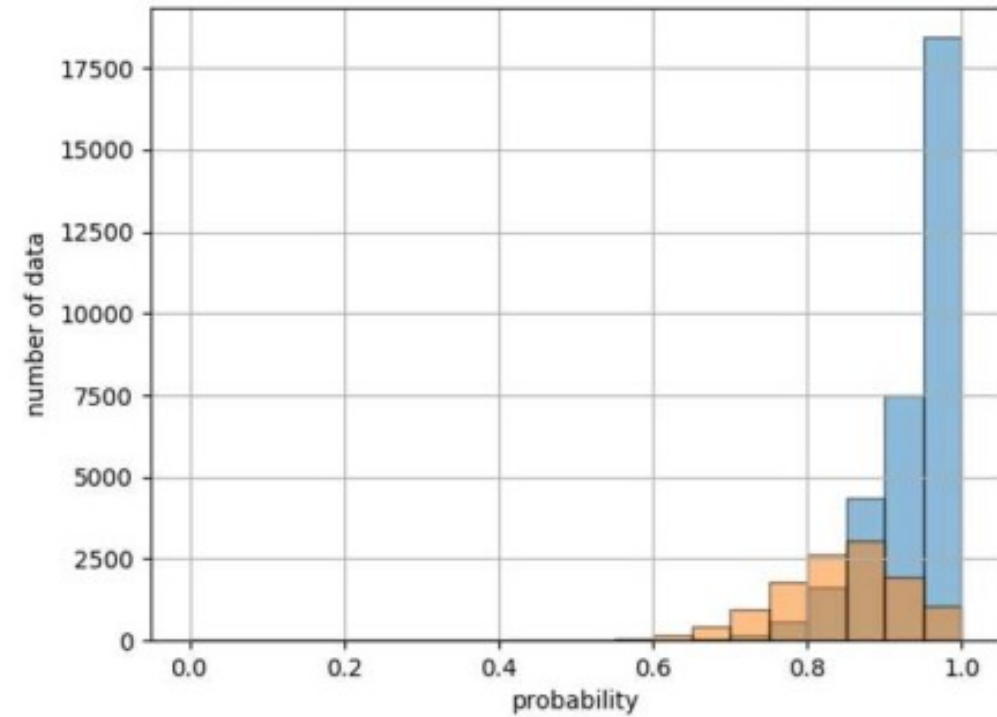
loss graph



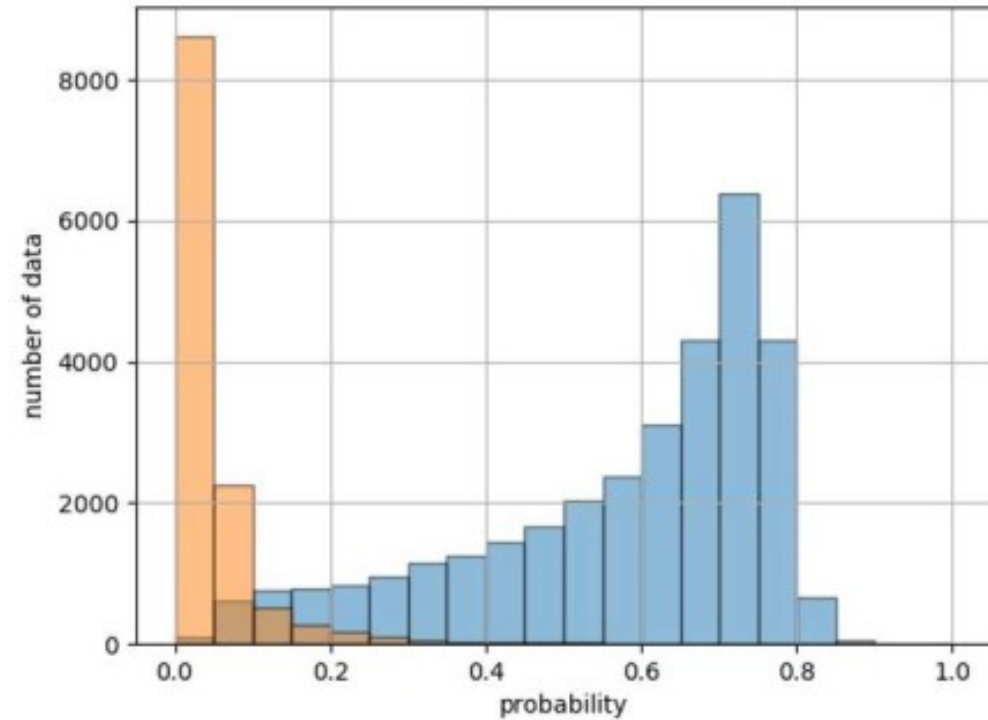
accuracy graph

- CNN trained with 30% symmetric-inclusive noise
 - randomly selecting the label from all classes
- All loss graphs shown are computed with PL-formula.
- Test data is clean!

CIFAR 10 training data with 30% symm-inc



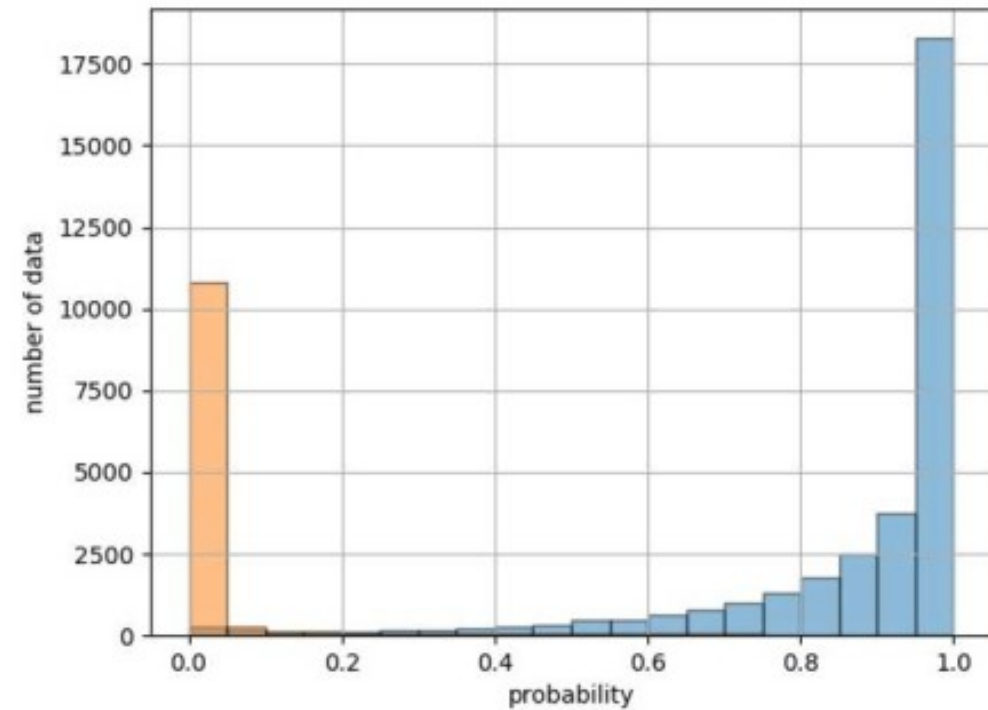
PL



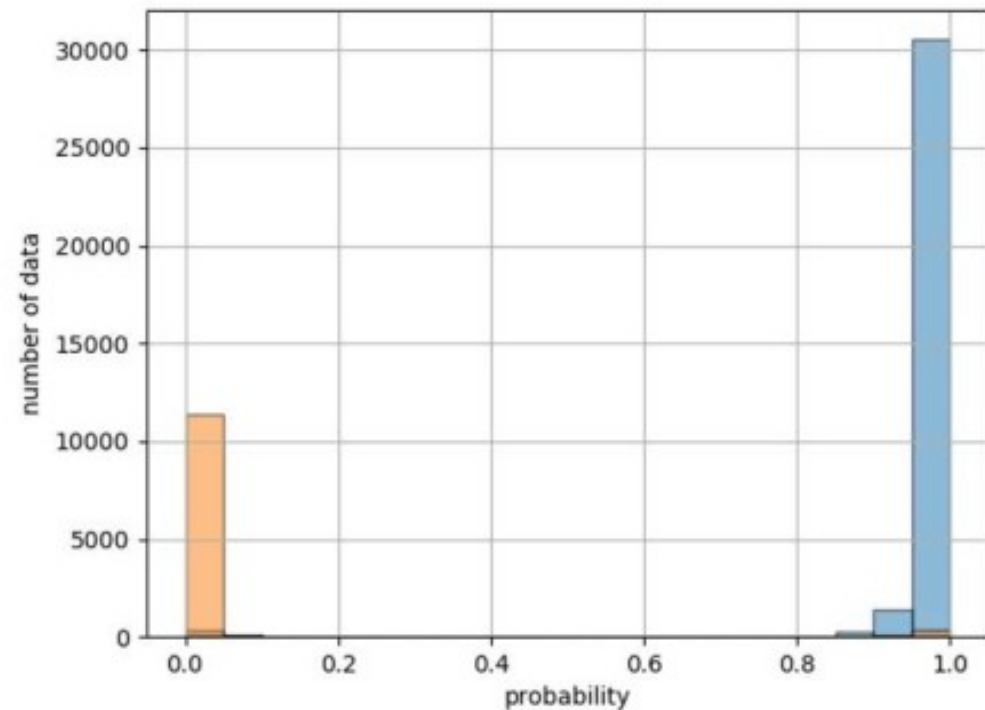
NL

- blue: correct labels; orange: wrong labels
 - PL causes the wrong labels to be considered correct.
 - NL causes the wrongly labeled data not to be considered as having the wrong label.
- CIFAR (Canadian Institute For Advanced Research) benchmark data set
10 classes (airplanes, cars, birds, cats, deer, dogs, frogs, horses, ships, trucks)

Next Steps: Selective NL and PL



NL->SelNL



NL->SelNL->SelPL

- SelNL: train only with data having confidence over $1/c$
- SelPL: train only with data having confidence over γ ($\sim$ clean data)
 - $\gamma = 0.5$
 - High confidence near 1 for almost all clean data

Overall Process of SelNLPL

▪ Algorithm

Input: Training data $(x, y) \in (\mathcal{X}, \mathcal{Y})$, network $f(x; \theta)$, total epoch T

for $i \leftarrow 1$ to T **do** ▷ NL

 Batch $\leftarrow$ Sample x

 Update f by minimizing Eq. 2

for $i \leftarrow 1$ to T **do** ▷ SelNL

 Batch $\leftarrow$ Sample x if $p_y > 1/c$

 Update f by minimizing Eq. 2

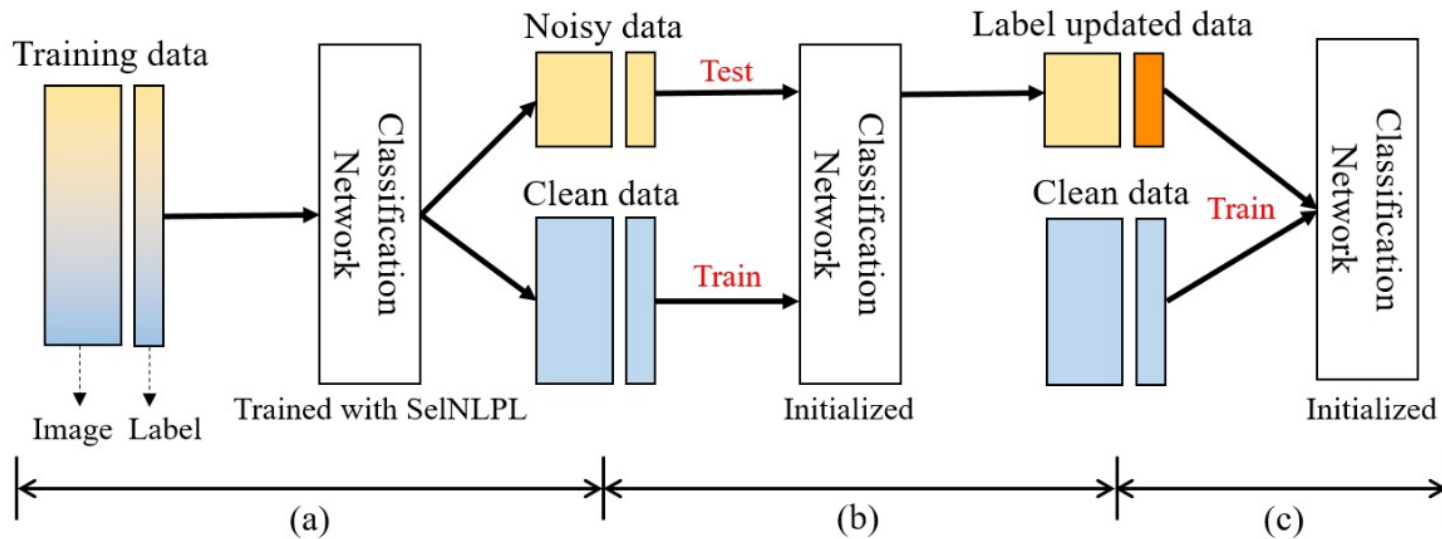
for $i \leftarrow 1$ to T **do** ▷ SelPL

 Batch $\leftarrow$ Sample x if $p_y > \gamma$

 Update f by minimizing Eq. 1

Output: Network $f(x; \theta)$

Pseudo Labeling for Semi-Supervised Learning



1. Divide training data into clean and noisy data using CNN trained with SelNLPL
2. Train CNN with clean data.
3. Label noisy data newly with soft label (not 1-hot) from CNN.
4. Train with clean and label-updated data.

Performance Comparison

Datasets	Model	Methods	<i>Symm</i>				<i>Asymm</i>			
			20	40	60	80	10	20	30	40
FashionMNIST	ResNet18	CE	93.24	92.09	90.29	86.20	94.06	93.72	92.72	89.82
		MAE [3]	80.39	79.30	82.41	74.73	74.03	63.03	58.14	56.04
		Forward T [25]	93.64	92.69	91.16	87.59	94.33	94.03	93.91	93.65
		Forward $\hat{T}$ [25]	93.26	92.24	90.54	85.57	94.09	93.66	93.52	88.53
		L_q [39]	93.35	92.58	91.30	88.01	93.51	93.24	92.21	89.53
		Truncated L_q [39]	93.21	92.60	91.56	88.33	93.53	93.36	92.76	91.62
		Ours	94.82	94.16	92.78	-	95.10	94.88	94.66	93.96
CIFAR10	ResNet34	CE	86.98	81.88	74.14	53.82	90.69	88.59	86.14	80.11
		MAE [3]	83.72	67.00	64.21	38.63	82.61	52.93	50.36	45.52
		Forward T [25]	88.63	85.07	79.12	64.30	91.32	90.35	89.25	88.12
		Forward $\hat{T}$ [25]	87.99	83.25	74.96	54.64	90.52	89.09	86.79	83.55
		L_q [39]	89.83	87.13	82.54	64.07	90.91	89.33	85.45	76.74
		Truncated L_q [39]	89.70	87.62	82.70	67.92	90.43	89.45	87.10	82.28
		Ours	94.23	92.43	88.32	-	94.57	93.35	91.80	89.86
CIFAR100	ResNet34	CE	58.72	48.20	37.41	18.10	66.54	59.20	51.40	42.74
		MAE [3]	15.80	9.03	7.74	3.76	13.38	11.50	8.91	8.20
		Forward T [25]	63.16	54.65	44.62	24.83	71.05	71.08	70.76	70.82
		Forward $\hat{T}$ [25]	39.19	31.05	19.12	8.99	45.96	42.46	38.13	34.44
		L_q [39]	66.81	61.77	53.16	29.16	68.36	66.59	61.45	47.22
		Truncated L_q [39]	67.61	62.64	54.04	29.60	68.86	66.59	61.87	47.66
		Ours	71.52	66.39	56.51	-	70.35	63.12	54.87	45.70

- Method achieves state of the art results.
- In case of asymmetric noise other methods seem better.
- But that is to some degree due to knowledge about asymmetry.
 - ➔ Knowledge about which class is mainly confused with which other class.

Weaker Than You Think

- Weakly-supervised learning is an ongoing research area.
- Recommendations for future WSL approaches:
 - Report how much WSL relies on presence of clean data.
 - Report how many cleanly annotated samples are required for a few-shot learning approach to reach the performance of a proposed WSL approach.
 - If thousands of weakly annotated samples are comparable to a handful of clean samples, WSL may not be a good choice.
 - If a proposed WSL method requires extra clean data (e.g. for validation), then non-WSL standard methods should be included in evaluation.